

Charles LeCroak

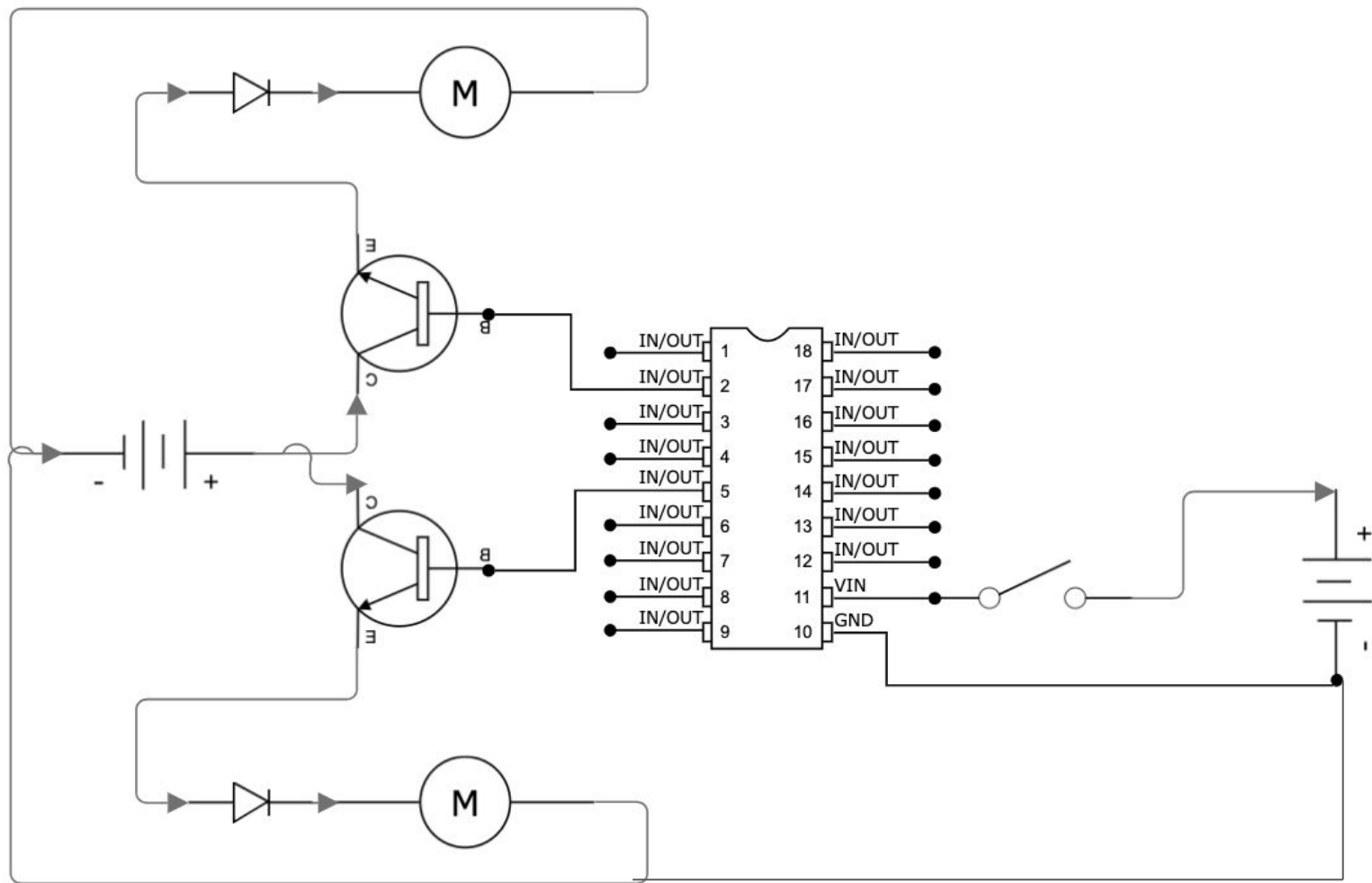
By Colin and Anshu

Intro

For our arduino project we built a remote control car using 2 ESP32 boards. They communicated using esp-now, a wireless communication protocol for ESP boards.

One ESP board hooks up to the computer with a joystick part. The other is in the car for the motors.

The circuit includes 2 5v motors, a 3.7v battery for the Arduino, and a 7.4V for the motors. We also have 2 switches that regulate the power from the batteries.



Receiver Code

```
8  #include <esp_now.h>
9  #include <WiFi.h>
10
11  static const int Motor1 = A4;
12  static const int Motor2 = A5;
13
14  int v1;
15  int v2;
16
17  // Structure example to receive data
18  // Must match the sender structure
19  typedef struct struct_message {
20      int v1;
21      int v2;
22  } struct_message;
23
24  // Create a struct_message called myData
25  struct_message myData;
26
```

```
27 // callback function that will be executed when data is received
28 void OnDataRecv(const uint8_t * mac, const uint8_t *incomingData, int len) {
29     memcpy(&myData, incomingData, sizeof(myData));
30
31     v1 = myData.v1;
32     v2 = myData.v2;
33
34     Serial.print("Bytes received: ");
35     Serial.println(len);
36     Serial.print("v1: ");
37     Serial.println(v1);
38     Serial.print("v2: ");
39     Serial.println(v2);
40     Serial.println();
41
42     analogWrite(Motor1, v1);
43     analogWrite(Motor2, v2);
44 }
```

```
46 void setup() {
47     // Initialize Serial Monitor
48     Serial.begin(9600);
49
50     // Set device as a Wi-Fi Station
51     WiFi.mode(WIFI_STA);
52
53     // Init ESP-NOW
54     if (esp_now_init() != ESP_OK) {
55         Serial.println("Error initializing ESP-NOW");
56         return;
57     }
58
59     // Once ESPNow is successfully Init, we will register for recv CB to
60     // get recv packer info
61     esp_now_register_recv_cb(esp_now_recv_cb_t(OnDataRecv));
62 }
```

```
/*
Rui Santos & Sara Santos - Random Nerd Tutorials
Complete project details at https://RandomNerdTutorials.com/esp-now-esp32-arduino-ide/
Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files.
The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.
*/

#include <esp_now.h>
#include <WiFi.h>

static const int Motor1 = A4;
static const int Motor2 = A5;

int v1;
int v2;

// Structure example to receive data
// Must match the sender structure
typedef struct struct_message {
  int v1;
  int v2;
} struct_message;

// Create a struct_message called myData
struct_message myData;

// callback function that will be executed when data is received
void OnDataRecv(const uint8_t * mac, const uint8_t *incomingData, int len) {
  memcpy(&myData, incomingData, sizeof(myData));

  v1 = myData.v1;
  v2 = myData.v2;

  Serial.print("Bytes received: ");
  Serial.println(len);
  Serial.print("v1: ");
  Serial.println(v1);
  Serial.print("v2: ");
  Serial.println(v2);
  Serial.println();

  analogWrite(Motor1, v1);
  analogWrite(Motor2, v2);
}

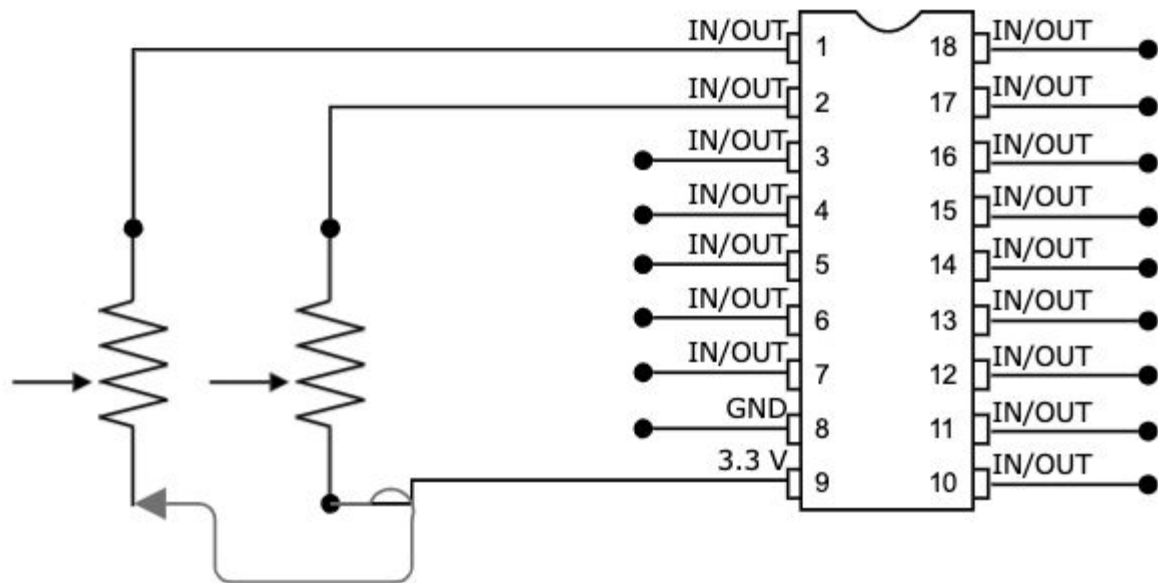
void setup() {
  // Initialize Serial Monitor
  Serial.begin(9600);
  // Set device as a Wi-Fi Station
  WiFi.mode(WIFI_STA);

  // Init ESP-NOW
  if (esp_now_init() != ESP_OK) {
    Serial.println("Error initializing ESP-NOW");
    return;
  }

  // Once ESPNow is successfully Init, we will register for recv CB to
  // get recv packer info
  esp_now_register_recv_cb(esp_now_recv_cb_t(OnDataRecv));
}

void loop() {
}

}
```



Controller Code

```
7  #include <WiFi.h>
8  #include <esp_now.h>
9
10 //reciever MAC address: c8:f0:9e:24:d4:5c
11 uint8_t broadcastAddress[] = {0xc8, 0xf0, 0x9e, 0x24, 0xd4, 0x5c};
12
13 static int xPIN = A7;
14 static int yPIN = A6;
15
16 // Structure example to send data
17 // Must match the receiver structure
18 typedef struct struct_message {
19     int v1;
20     int v2;
21 } struct_message;
22
23 // Create a struct_message called myData
24 struct_message myData;
25
26 esp_now_peer_info_t peerInfo;
```

```
28 // callback when data is sent
29 void OnDataSent(const uint8_t *mac_addr, esp_now_send_status_t status) {
30     Serial.print("\r\nLast Packet Send Status:\t");
31     Serial.println(status == ESP_NOW_SEND_SUCCESS ? "Delivery Success" : "Delivery Fail");
32 }
```

```
34 void setup() {
35     // Init Serial Monitor
36     Serial.begin(9600);
37
38     pinMode(xPIN, INPUT);
39     pinMode(yPIN, INPUT);
40
41     // Set device as a Wi-Fi Station
42     WiFi.mode(WIFI_STA);
43
44     // Init ESP-NOW
45     if (esp_now_init() != ESP_OK) {
46         Serial.println("Error initializing ESP-NOW");
47         return;
48     }
49
50     // Once ESPNow is successfully Init, we will register for Send CB to
51     // get the status of Trasnmitted packet
52     esp_now_register_send_cb(OnDataSent);
53
54     // Register peer
55     memcpy(peerInfo.peer_addr, broadcastAddress, 6);
56     peerInfo.channel = 0;
57     peerInfo.encrypt = false;
58
59     // Add peer
60     if (esp_now_add_peer(&peerInfo) != ESP_OK){
61         Serial.println("Failed to add peer");
62         return;
63     }
64 }
```

```
66 void loop() {
67     int xAxis = analogRead(xPIN);
68     int yAxis = analogRead(yPIN);
69
70     xAxis = (int) (255*(xAxis-1907)/1907);
71     yAxis = (int) (255*(yAxis-1907)/1907);
72
73     xAxis = (xAxis > 255) ? 255 : xAxis;
74     yAxis = (yAxis > 255) ? 255 : yAxis;
75
76     int m1 = xAxis - yAxis;
77     int m2 = xAxis + yAxis;
78
79     // Set values to send
80     myData.v1 = m2;
81     myData.v2 = m1;
82
83     // Send message via ESP-NOW
84     esp_err_t result = esp_now_send(broadcastAddress, (uint8_t *) &myData, sizeof(myData));
85
86     if (result == ESP_OK) {
87         Serial.println("Sent with success");
88     }
89     else {
90         Serial.println("Error sending the data");
91     }
92     delay(100);
93 }
```

Full Source & Demo Datas - Random Street Tutorial
Complete project details at <https://randomstreetutorials.com/esp-now-esp32-arduino-106/>
Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files.
The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

```
*/
// Full Source & Demo Datas - Random Street Tutorial
// Complete project details at https://randomstreetutorials.com/esp-now-esp32-arduino-106/
// Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files.
// The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.
*/
#include <ESP32.h>
#include <esp_now.h>

//Receiver MAC address: 08:00:2e:14:04:1c
uint8_t broadcastAddress[] = {0x08, 0x00, 0x2e, 0x14, 0x04, 0x1c};

static int sPort = 81;
static int yPort = 82;

// Structure example to send data
// Must match the receiver structure
typedef struct struct_message {
  int v1;
  int v2;
} struct_message;

// Create a struct_message called myData
struct_message myData;

esp_now_peer_info_t peerInfo;

// Callback when data is sent
void onDataSent(const uint8_t *mac_addr, esp_now_send_status_t status) {
  Serial.println("Successful Packet Sent Status:");
  Serial.println(mac_addr == ESP_NOW_BROADCAST ? "Delivery Success" : "Delivery Fail");
}

void setup() {
  // Init Serial Monitor
  Serial.begin(9600);

  pinMode(LED, OUTPUT);
  pinMode(yPort, OUTPUT);
  // Set Device as a Wi-Fi Station
  WiFi.mode(WIFI_STA);

  // Init ESP-NOW
  if (esp_now_init()) != ESP_OK {
    Serial.println("Error Initializing ESP-NOW");
    return;
  }

  // Once ESPNow is successfully Init, we will register for Send CB to
  // get the status of Transmitted packet
  esp_now_register_send_cb(onDataSent);

  // Register peer
  memcpy(&peerInfo, peer_addr, broadcastAddress, 6);
  peerInfo.channel = 0;
  peerInfo.encrypt = false;
  // Add peer
  if (esp_now_add_peer(&peerInfo) != ESP_OK) {
    Serial.println("Failed to add peer");
    return;
  }
}

void loop() {
  int xData = analogRead(xPort);
  int yData = analogRead(yPort);

  xData = (int) (255*(xData-107)/107);
  yData = (int) (255*(yData-107)/107);

  xData = (xData > 255) ? 255 : xData;
  yData = (yData > 255) ? 255 : yData;

  int x1 = xData - yData;
  int x2 = xData + yData;

  // Set values to send
  myData.v1 = x2;
  myData.v2 = x1;
  // Send message via ESP-NOW
  esp_err_t result = esp_now_send(broadcastAddress, (uint8_t *) &myData, sizeof(myData));

  if (result == ESP_OK) {
    Serial.println("Sent with success");
  }
  else {
    Serial.println("Error sending the data");
  }
  delay(100);
}
```

Resources

<https://randomnerdtutorials.com/getting-started-with-esp32/>

<https://www.amazon.com/Teyleten-Robot-ESP-WROOM-32-Development-Microcontroller/dp/B08246MCL5?th=1>

<https://randomnerdtutorials.com/esp32-web-server-arduino-ide/>

<https://randomnerdtutorials.com/esp32-servo-motor-web-server-arduino-ide/>

<https://lastminuteengineers.com/creating-esp32-web-server-arduino-ide/>

<https://randomnerdtutorials.com/esp32-access-point-ap-web-server/>

<https://randomnerdtutorials.com/esp32-servo-motor-web-server-arduino-ide/>

<https://developer.mozilla.org/en-US/docs/Web/API/XMLHttpRequest/open>

https://github.com/mattsroufe/esp32_rc_cars